

You are the senior full-stack engineer for PSM OS.

I am attaching:

- Equipment Registry mockup screenshot
- Phase 1 file structure
- Equipment Registry requirements

Task: Fix and complete the Equipment Registry Detail Page starting from the Overview tab. The current UI looks good, but many items are static or not working. Make everything real, API-driven, saved to database, and instantly refreshed in the UI.

Do not redesign the page. Keep the current dark industrial UI and layout. Make it functional.

Build backend first, then frontend.

Tech stack:

- Backend: NestJS + Prisma + PostgreSQL
- Frontend: Next.js 14 + TypeScript + Tailwind + shadcn/ui + React Query + React Hook Form + Zod
- Storage: S3 or Supabase Storage
- QR: server-side QR generation

Main bugs to fix: 1. QR code is not generated in real time. 2. Edit Equipment opens but does not properly save and refresh all cards. 3. Notes save but do not refresh correctly. 4. Document upload saves but preview/list is not updated properly. 5. Quick Actions are visible but not all functional. 6. Linked record counters are static/mock data. 7. Tabs exist but many do not show real database data.

Required tabs and exact behavior:

1. Overview Show live API data:
2. Equipment header
3. Equipment photo
4. Tag, name, status, type
5. Location breadcrumbs
6. QR code
7. Manufacturer, model, serial number, fabrication year
8. Design Conditions card
9. Operating Conditions card
10. Status & Criticality card
11. Classification card
12. Fluid & Hazard Info card
13. Safety Information card
14. Inspection Overview card

15. Equipment Hierarchy preview
16. Description
17. Cross-Module Summary
18. Linked Records preview
19. Notes preview
20. Quick Actions

Overview Quick Actions must work:

- Edit Equipment: open modal/form, save to DB, refetch equipment detail, update all cards.
- Generate QR Label: call API, generate QR, save QR record/file, refresh QR display.
- Add Child Equipment: open child equipment form, save parent-child relationship, refresh hierarchy.
- Upload Document: upload file, create equipment document relation, show preview/list instantly.
- View Linked Records: open Linked Records tab.
- Inspection History: open Inspection tab.

- Documents: open Documents tab.

- Details Create full editable details view/form:
 - tag

 - name
 - description
 - type
 - status
 - manufacturer
 - model
 - serial number
 - commission date
 - fabrication year
 - site/unit/area/system
 - design pressure/temp
 - operating pressure/temp
 - phase
 - fluid service
 - hazard class
 - SDS reference
 - exposure limits
 - environmental impact
 - criticality
 - safety critical
 - equipment classification

 - area classification

- Documents Show controlled equipment documents:

- P&ID
- datasheet
- vendor manual
- SOP
- certificate
- inspection report
- other Features:
- upload
- preview
- download
- replace version
- delete/unlink
- show version, uploaded by, uploaded date After upload, list must refresh immediately.
- Linked Records Show real linked records grouped by:
- PTW
- MOC
- HAZOP
- PSSR
- Actions
- Incidents
- Documents Each cross-module summary tile on Overview must navigate here with filter applied.
- History Timeline Show real timeline events:
- equipment created
- equipment edited
- QR generated
- document uploaded
- note added
- inspection added
- child equipment added
- linked record created
- status changed Every create/update/upload/delete must write a timeline event.
- Inspection Create inspection CRUD:
- add inspection

- edit inspection
- delete inspection
- upload inspection attachments Fields:
 - inspection type
 - inspection date
 - inspector
 - result: Pass / Monitor / Fail
 - observation
 - next inspection date
 - RBI priority
- attachments Overview Inspection card must use latest inspection and next inspection date.
- Hierarchy Show clickable parent-child equipment tree. Add Child Equipment must create child record and refresh tree. Clicking child equipment opens that equipment detail page.
- Attachments For non-controlled files:
 - equipment photos
 - nameplate photos
 - field evidence
 - misc files Features:
 - upload
 - preview thumbnail
 - download
 - delete
- Notes Make notes fully working:
 - add note
 - edit note
 - delete note
 - show author
 - show timestamp
 - refetch after save
 - show latest notes on Overview

Required backend additions if missing:

- equipment_qr_codes
- equipment_documents
- equipment_attachments
- equipment_notes
- equipment_inspections
- equipment_timeline_events
- equipment_relationships
- equipment_linked_records

Required API endpoints:

- GET /equipment/:id
- PATCH /equipment/:id
- POST /equipment/:id/qr
- GET /equipment/:id/qr
- POST /equipment/:id/documents
- GET /equipment/:id/documents
- POST /equipment/:id/attachments
- GET /equipment/:id/attachments
- POST /equipment/:id/notes
- PATCH /equipment/:id/notes/:noteId
- DELETE /equipment/:id/notes/:noteId
- GET /equipment/:id/linked-records
- GET /equipment/:id/timeline
- POST /equipment/:id/inspections
- GET /equipment/:id/inspections
- PATCH /equipment/:id/inspections/:inspectionId
- DELETE /equipment/:id/inspections/:inspectionId
- POST /equipment/:id/children
- GET /equipment/:id/hierarchy
- GET /equipment/:id/summary

Rules:

- No mock data except seed data.
- All data must come from backend APIs.
- Use React Query refetch/invalidate after every mutation.
- Add loading, empty, and error states.
- Add permission checks for edit, upload, delete, QR generation.
- Add audit logs for every mutation.
- Add timeline event for every mutation.
- Keep file structure aligned with Phase 1 structure.
- If a needed file/component/API is missing from the file structure, create it in the correct folder.

Start with Overview tab first. Make Overview fully working before implementing the other tabs.

Final output: 1. Files created/changed. 2. Backend changes. 3. Frontend changes. 4. API endpoints added. 5. Bugs fixed. 6. What still remains.